NeuroLink 3.0: A Privacy-Preserving N-of-1 Deep Autoencoder for Longitudinal Behavioral Drift Detection in Keystroke Dynamics

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Abstract

We present NeuroLink 3.0, a privacy-preserving N-of-1 deep autoencoder for detecting longitudinal behavioral drift in keystroke dynamics. Unlike population-based approaches, our system learns a personalized baseline from an individual's own typing history and flags sessions that deviate from it using reconstruction error. Trained on 42,501 keybr.com typing lessons spanning 17 months, our undercomplete autoencoder (12→32→16→ ***10****→16→32→12, 2,230 parameters) achieves a variance-weighted validation* R^2 *of* ***0.9586*** *under differential privacy (*ε=7.9983*,* δ=10^{-5}*) via DP-SGD. We introduce* residual *Execution Velocity (EV) and Error-Speed Recovery Penalty (ESRP) features — temporal deviation signals that resolve a structural reconstruction ceiling present in rolling-mean formulations, raising EV* R^2 *from 0.234 to 0.954 and ESRP* R^2 *from 0.113 to 0.854. A bootstrap confidence interval derived from 2,000 resamples of the training Longitudinal Drift Index (LDI) enables three-tier anomaly alerting (HIGH / MARGINAL / NORMAL), and an empirically calibrated Ghost V3 predictor derives its drift-adjustment scale factor directly from data rather than relying on a hand-tuned constant.*

1. Introduction

Behavioral biometrics — the study of *how* individuals interact with digital devices — offers a passive, non-invasive window into cognitive and motor state [11]. Among the richest behavioral signals is *keystroke dynamics*: the millisecond-level rhythm, speed, error patterns, and per-key latency produced during typing. These signals capture aspects of neuromuscular coordination that may reflect fatigue, stress, or longer-term behavioral change, making them attractive for longitudinal personal monitoring.

Existing keystroke dynamics systems predominantly focus on *authentication*, comparing a user's current pattern against a stored population template to verify identity [8]. This framing treats the typing template as static and optimizes for *inter-person* discrimination. A complementary and less-studied problem is detecting *within-person behavioral drift* over time: has this individual's typing changed relative to their own established baseline? This question is directly relevant to cognitive health monitoring, workplace fatigue detection, and human-computer interaction research, but standard population-norm approaches fail at it by design — a consistently slow typist would be permanently flagged as anomalous.

The N-of-1 Paradigm. The N-of-1 clinical trial design [9] treats each individual as their own control. Applying this philosophy to deep learning, we train a model *exclusively* on a single user's longitudinal data, learning what regularity looks like *for that person*, and flagging deviations from their own history. This avoids all population-norm assumptions and is inherently personalized.

NeuroLink 3.0. We present NeuroLink 3.0, a system that learns a compact 10-dimensional *Muscle Memory Signature* from 42,501 personal typing lessons using an undercomplete autoencoder. High reconstruction error on a new lesson signals behavioral deviation. This per-lesson error, aggregated into a 7-day rolling Longitudinal Drift Index (LDI), forms the core anomaly signal. Differential privacy via DP-SGD [1] ensures the model cannot memorize individual lessons.

Contributions. This paper makes the following contributions:

1. **N-of-1 undercomplete autoencoder** trained on 17 months of personal keystroke data with a 10-dimensional bottleneck that learns individual behavioral regularity without reference to population norms.
2. **Residual feature engineering** for EV and ESRP that resolves a structural reconstruction ceiling caused by using rolling-mean formulations — features that cannot be reconstructed from a single-lesson snapshot at inference

time.

3. **Differential privacy** via Opacus DP-SGD [14] with RDP accounting, achieving ε=7.9983 at δ=10^{-5} — providing a formal privacy guarantee that the behavioral baseline cannot leak raw lesson data.
4. **Bootstrap confidence interval alerting**: 2,000 bootstrap resamples of training LDI values produce a 95% CI band enabling three-tier classification (HIGH / MARGINAL / NORMAL), reducing false positives compared to a fixed point threshold.
5. **Empirical Ghost V3**: a drift-adjusted typing pace predictor whose scale factor is derived from a linear regression of daily LDI ratio against observed WPM across the full 42,501-lesson dataset, replacing a hand-tuned constant.

2. Related Work

Keystroke Dynamics. Keystroke dynamics as a behavioral biometric were first studied for authentication in the 1980s [11]. Killourhy and Maxion [8] established a benchmark dataset and systematic comparison of 14 classifiers, finding that simple distance-based methods often match more complex approaches. Subsequent work introduced deep learning for keystroke authentication, including RNNs for sequence modeling and Siamese networks for one-shot verification [10]. All of these approaches optimize for *cross-user* discrimination. Our work targets the orthogonal goal of *within-user* longitudinal drift detection, for which population-template methods are fundamentally ill-suited.

Anomaly Detection with Autoencoders. Autoencoders were introduced for nonlinear dimensionality reduction by Rumelhart *et al*. [12] and popularized for deep representations by Hinton and Salakhutdinov [7]. Their application to anomaly detection exploits the observation that a model trained exclusively on normal data will reconstruct normal inputs well but yield high error on anomalous inputs [2]. Vincent *et al*. [13] introduced denoising autoencoders, training to reconstruct clean inputs from corrupted versions, which improves robustness and forces learning of more meaningful representations. We adopt a denoising variant with reproducible 10% input masking.

Differential Privacy in Deep Learning. Dwork *et al*. [5] formalized differential privacy (DP) as a mathematically rigorous definition of privacy for statistical algorithms. Abadi *et al*. [1] extended DP to neural network training via DP-SGD: gradient clipping followed by calibrated Gaussian noise addition, with privacy budget tracked via moments accountant. The Opacus library [14] provides an efficient PyTorch implementation with Rényi Differential Privacy (RDP) accounting. In the context of personal behavioral data — where lessons encode daily behavioral patterns — formal privacy guarantees are especially important.

Longitudinal Behavioral Monitoring. Digital biomarker research has demonstrated that passive sensing of device-interaction patterns can detect cognitive fluctuations [4]. Typing-based monitoring has been proposed for Parkinson's disease [6] and attention deficit detection [3]. These systems typically rely on engineered statistical features rather than learned representations. Our work contributes a learned, privacy-preserving representation that adapts entirely to the individual without supervision.

N-of-1 Designs. N-of-1 randomized controlled trials [9] treat each participant as their own control across multiple crossover periods, enabling causal inference at the individual level. The philosophy has been applied to personalized medicine and digital therapeutics. Our work is, to our knowledge, the first application of the N-of-1 design philosophy to a deep autoencoder for keystroke behavioral monitoring.

3. Method

3.1. Dataset

Data were collected from keybr.com, an adaptive typing practice platform, by a single participant over 17 months (December 2024 – May 2026). Each lesson record contains: UTC timestamp, overall typing speed (characters per minute), error count, session duration (ms), session length (characters), and per-key average keystroke time and error count for 27 characters (a–z and space). After timestamp parsing and WPM conversion (WPM = cpm/5), the dataset contains **42,501 lessons** with a mean speed of 87.6 WPM and a peak of 136.6 WPM. No personally identifiable information beyond typing behavior is present.

3.2. Feature Engineering

We extract 12 behavioral features organized into three groups.

Speed and Variance Features. *Residual Execution Velocity* (**EV**) captures per-lesson speed surprise relative to recent trend:

$$\mathrm{EV}_t = \mathrm{WPM}_t - \frac{1}{7}\sum_{i=0}^{6} \mathrm{WPM}_{t-i} \tag{1}$$

Positive EV indicates a faster-than-trend session; negative indicates drift. *Rolling Stability Variance* (**RSV**) is the 30-lesson rolling standard deviation of WPM, measuring consistency over time. *WPM vs. 50-lesson rolling* (**wpm_vs_50roll**) normalizes current WPM against the long-term trend.

Error Features. *Residual Error-Speed Recovery Penalty* (**ESRP**) captures per-lesson error surprise:

$$\text{ESRP}_t = r_t \cdot \frac{1000}{\text{WPM}_t} - \frac{1}{5}\sum_{i=0}^{4}\left(r_{t-i} \cdot \frac{1000}{\text{WPM}_{t-i}}\right) \quad (2)$$

where $r_t = \text{errors}_t/\text{chars}_t$ is the per-lesson error rate. *Error Entropy* (**error_entropy**) is the Shannon entropy of the per-key error count distribution, measuring whether errors are concentrated on specific keys or spread uniformly.

Structural Features. *Sequential Stamina Fatigue* (**SSF**) encodes intra-day fatigue as the product of the lesson's position within the day and the within-day WPM z-score. *Circadian Temporal Anchors* (**CTA**, **CTA_cos**) are the sine and cosine of the session's UTC hour, encoding time-of-day in a continuous circular representation. *Lesson Duration* (**lesson_dur_z**) is $\log_1(session_time)$, log-transforming the heavy-tailed duration distribution. *Active Character Fraction* (**active_char_frac**) is the proportion of the 27-character vocabulary present in the lesson, encoding curriculum depth. *Hand Asymmetry* (**hand_asymmetry**) is the ratio of mean left-hand to right-hand key latency across available keys, capturing motor lateralization. *Practice Gap* (**practice_gap**) is $\log_1(h)$ where h is the elapsed hours since the previous lesson (clipped to 72h), encoding schedule regularity.

The Residual Feature Insight. Prior work defined EV as the 7-lesson rolling mean WPM. This creates a *structural reconstruction ceiling*: at inference time, the model receives a single lesson's feature vector. Reconstructing a rolling mean requires access to the 6 preceding lessons — information unavailable in the bottleneck. The autoencoder therefore cannot learn to reconstruct rolling-mean EV from single-lesson features, regardless of capacity or training time.

By reformulating as the *deviation* from the rolling mean (Eqs. 1–2), EV and ESRP become single-lesson quantities that *are* predictable from other within-lesson features (WPM magnitude, error rate, circadian time). This yields a theoretically principled formulation that is empirically validated in Section 4.2.

3.3. Architecture

We use a symmetric undercomplete autoencoder:

$$12 \to 32 \to 16 \to \mathbf{10} \to 16 \to 32 \to 12$$

with 2,230 trainable parameters. The encoder uses LeakyReLU activations (α=0.01); the 10-dimensional bottleneck uses Tanh to bound the *Muscle Memory Signature*. The decoder mirrors the encoder with LeakyReLU activations and a linear output layer. All weights are initialized with Xavier uniform initialization. The bottleneck compression ratio (10/12 = 0.83) was selected to maintain sufficient capacity for 12 features while still forcing compression.

A *reproducible denoising mask* applies 10% random feature zeroing during training using a deterministic random state seeded by epoch and batch index, ensuring reproducibility under DP noise.

3.4. Differentially Private Training

We train with DP-SGD via the Opacus library [14] using the RDP accountant. Hyperparameters: target ε=8.0, δ=10^{-5}, max gradient norm C=1.0, batch size 64, yielding noise multiplier σ=0.5994. The final privacy expenditure is ε=7.9983 at δ=10^{-5}.

The weighted MSE loss emphasizes features with diagnostic importance:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{w}^\top(\hat{\mathbf{x}}_i - \mathbf{x}_i)^2 \quad (3)$$

where $\mathbf{w} \in \mathbb{R}^{12}$ is a fixed weight vector. Weights are set to 4.0 for lesson_dur_z (hardest to reconstruct), 2.5 for practice_gap, 2.0 for RSV, hand_asymmetry, and error_entropy, and 1.0 for EV and ESRP (residuals are structurally easy to predict).

We use the Adam optimizer with initial learning rate 2×10^{-3} and `ReduceLROnPlateau` scheduling (factor = 0.5, patience = 5), which halves the learning rate when validation loss stagnates. This allows the model to escape early local minima under DP noise by adaptively reducing gradient step size, in contrast to cosine annealing which fixes the decay schedule. Training uses a patience of 20 epochs for early stopping with best-checkpoint restoration.

3.5. Longitudinal Drift Index and Anomaly Detection

For each lesson, per-lesson reconstruction MSE is computed as:

$$\ell_t = \frac{1}{12}\sum_{j=1}^{12}(\hat{x}_{t,j} - x_{t,j})^2$$

Daily LDI is the mean per-lesson MSE across all lessons in a calendar day, smoothed with a 7-day rolling mean to suppress lesson-level noise:

$$\text{LDI}_{7\text{d}}(d) = \frac{1}{7}\sum_{k=0}^{6}\bar{\ell}_{d-k}$$

Bootstrap Confidence Interval Alerting. Rather than a single point threshold, we estimate the uncertainty in the training-period anomaly threshold. We draw 2,000 bootstrap resamples of training-period LDI values and compute

the $\mu+2\sigma$ threshold for each resample. The 2.5th and 97.5th percentiles of these thresholds define the lower and upper CI bounds $(\theta_{\text{lo}}, \theta_{\text{hi}})$. Each day is then classified as:

$$\text{tier}(d) = \begin{cases} \text{HIGH} & \text{if } \text{LDI}_{7\text{d}}(d) \geq \theta_{\text{hi}} \\ \text{MARGINAL} & \text{if } \theta_{\text{lo}} \leq \text{LDI}_{7\text{d}}(d) < \theta_{\text{hi}} \\ \text{NORMAL} & \text{if } \text{LDI}_{7\text{d}}(d) < \theta_{\text{lo}} \end{cases}$$

HIGH days represent confident anomalies; MARGINAL days fall within the threshold uncertainty band and warrant monitoring but not immediate action.

4. Experiments

4.1. Experimental Setup

Data Split. We use a strictly chronological split to prevent temporal leakage: 75% training (31,875 lessons, through Feb 2026), 10% validation (4,250 lessons, Feb–Mar 2026), 15% test (6,376 lessons, Apr–May 2026). Features are normalized with a `StandardScaler` fit on the training set only and clipped to $[-4, 4]$ sigma to suppress outliers.

Metric. We report variance-weighted R^2 across all 12 features as the primary metric, computed on the validation and test sets. Per-feature R^2 quantifies each feature's individual reconstruction quality, enabling diagnosis of bottleneck capacity limitations. For anomaly detection we report the LDI statistics, bootstrap CI bounds, and tier-wise day counts.

4.2. Ablation: Residual vs. Rolling-Mean Features

Table 1 compares three configurations on the same data and architecture to isolate the effect of residual feature engineering. *v4* uses rolling-mean EV and ESRP; *v5* introduces residual formulations but with a high initial learning rate (5×10^{-3}) and cosine annealing; *v5.1* adds ReduceLROnPlateau scheduling and rebalanced feature weights.

Version	Val R^2	EV R^2	ESRP R^2	Gap R^2	Epoch
v4 (rolling mean)	0.9502	0.234	0.113	0.411	55
v5 (residual)	0.8860	0.948	0.730	0.000	10
v5.1 (ours)	**0.9586**	**0.954**	**0.854**	**0.904**	**54**

Table 1. Ablation study across three versions. v5 demonstrates that residual features dramatically improve EV and ESRP R^2 but suffer from premature convergence. v5.1 recovers full performance by combining residual features with adaptive learning rate scheduling and weight rebalancing, achieving best results on all metrics simultaneously.

The v5 result reveals an important failure mode: residual EV and ESRP, being structurally easier to predict, cause the model to converge rapidly to a local minimum (best epoch 10 vs. 55 in v4). The high initial learning rate (5×10^{-3}) combined with DP noise then degrades harder features (lesson_dur_z collapses to $R^2 = -0.488$). ReduceLROnPlateau with factor 0.5 halves the learning rate at epoch 24, allowing sustained improvement through epoch 54 and recovering all features.

4.3. Main Results

Table 2 reports per-feature validation R^2 for NeuroLink 3.0 v5.1. The model achieves $R^2 \geq 0.90$ on 10 of 12 features.

Feature	Description	Val R^2
EV*	Residual speed surprise	0.954
RSV	30-lesson WPM std	0.964
SSF	Intra-day fatigue	0.980
ESRP*	Residual error surprise	0.854
CTA	Circadian sin	0.940
wpm_vs_50roll	Long-term WPM trend	0.890
CTA_cos	Circadian cos	0.981
lesson_dur_z	Session duration (log)	0.827
active_char_frac	Curriculum depth	0.954
hand_asymmetry	L/R latency ratio	0.990
practice_gap	log(hours since last lesson)	0.904
error_entropy	Shannon entropy of errors	0.989
Variance-weighted overall		**0.9586**
Test R^2		**0.9604**

Table 2. Per-feature validation R^2 for NeuroLink 3.0 v5.1. Features marked * use the residual formulation introduced in v5. The model achieves $R^2 \geq 0.90$ on 10 of 12 features under differential privacy (ε=7.9983, δ=10^{-5}).

Training Dynamics. The model trains for all 60 epochs, using the full DP budget (ε=7.9983). `ReduceLROnPlateau` reduces LR from 2×10^{-3} to 1×10^{-3} at epoch 24, after which validation loss decreases monotonically from 0.059 to 0.034 at the best checkpoint (epoch 54).

4.4. Anomaly Detection Results

Table 3 reports LDI statistics and tier-wise day counts over 435 deployment days.

The anomaly explainability analysis identifies the dominant feature contributors to each HIGH anomaly day. The top HIGH anomaly day (LDI = 0.4102, Feb 28, 2025) is driven primarily by wpm_vs_50roll (18%), error_entropy (13%), and CTA_cos (13%), suggesting an atypical combination of speed deviation, error concentration, and unusual session timing.

Statistic	Value
LDI mean (μ)	0.05480
LDI std (σ)	0.07583
Point threshold ($\mu + 2\sigma$)	0.20646
Bootstrap 95% CI lower (θ_{lo})	0.17443
Bootstrap 95% CI upper (θ_{hi})	0.23522
HIGH days ($\geq \theta_{\text{hi}}$)	15 (3.4%)
MARGINAL days (CI band)	21 (4.8%)
NORMAL days ($< \theta_{\text{lo}}$)	399 (91.7%)

Table 3. LDI statistics and three-tier anomaly classification over 435 deployment days. HIGH days represent confident anomalies (above the upper 95% CI bound). MARGINAL days fall within the threshold uncertainty band.

4.5. Empirical Ghost V3 Calibration

We fit a linear regression of mean daily WPM against LDI ratio (current LDI / baseline LDI):

$$\overline{\text{WPM}}_d = a + b \cdot \frac{\text{LDI}_{7\text{d}}(d)}{\mu_{\text{LDI}}}$$

The empirically derived slope b yields a scale factor:

$$s_{\text{empirical}} = \text{clip}\left(\frac{-b}{\overline{\text{WPM}}},\ 0.01,\ 0.15\right)$$

The data-driven scale factor replaces the arbitrary constant 0.05 used in prior versions, directly quantifying the fractional WPM reduction per unit of normalized drift. A drift-adjusted session completion time is then predicted as:

$$T_{\text{Ghost V3}} = T_{\text{base}} \cdot (1 + (r - 1) \cdot s_{\text{empirical}})$$

where $r = \text{LDI}_{\text{current}}/\mu_{\text{LDI}}$ and T_{base} is estimated from the user's recent WPM history.

5. Conclusion

We presented NeuroLink 3.0, a privacy-preserving N-of-1 deep autoencoder for longitudinal behavioral drift detection in keystroke dynamics. Trained on 42,501 personal typing lessons spanning 17 months, the system achieves a variance-weighted validation R^2 of 0.9586 under formal differential privacy (ε=7.9983, δ=10^{-5}).

Our primary technical contribution is the identification and resolution of a *structural reconstruction ceiling* in temporal behavioral features. By reformulating rolling-mean EV and ESRP as per-lesson residuals — deviations from the rolling mean — we convert temporally-aggregated features that are structurally impossible to reconstruct from a single-lesson snapshot into individually predictable surprise signals. This raises EV R^2 from 0.234 to 0.954 and ESRP R^2 from 0.113 to 0.854, demonstrating that the right feature formulation matters as much as architecture or training procedure.

The combination of `ReduceLROnPlateau` scheduling with rebalanced feature weights recovers overall reconstruction quality (Val R^2 0.9586 vs. 0.9502 in v4) while retaining the residual feature benefits — achieving the best results across all metrics simultaneously.

The bootstrap confidence interval alerting framework provides statistically grounded three-tier classifications (15 HIGH, 21 MARGINAL, 399 NORMAL days over 435 deployment days), and the empirically calibrated Ghost V3 predictor derives its drift-adjustment scale factor directly from the observed LDI–WPM relationship across the full dataset.

Limitations and Future Work. The system is trained on a single subject's data from a structured typing practice platform. Ground-truth validation against externally recorded behavioral events (illness, sleep disruption, high-stress periods) has not been conducted. Future work should: (1) recruit multiple participants to evaluate cross-person generalizability of the N-of-1 design; (2) collect contemporaneous behavioral annotations to validate whether HIGH anomaly days correspond to subjectively meaningful events; (3) integrate real-time keystroke capture from natural typing environments; and (4) compare against simpler baselines (e.g., rolling WPM z-score thresholding) to formally justify the autoencoder's added complexity. Despite these limitations, NeuroLink 3.0 demonstrates that a compact, 2,230-parameter network can learn a stable, individually personalized representation of human typing behavior that is resilient to DP noise and sensitive to longitudinal behavioral change.

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